General Bites

A strategically scoped General Class upgrade program for technicians, designed to fit in 12 monthly club meetings.

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What This Is

General Bites is the sibling of Extra Bites, retargeted for the General Class upgrade. It is **not** a class. It is a 12-month sequence of short club-meeting concept lessons plus printable Self-study reference · Practice exams at eham.net/exams sheets, designed to move a motivated Technician licensee through the General Class exam in one calendar year.

The system uses **strategic scoping**: it deliberately does not try to teach every question in the 422-question General pool. It teaches the smallest realistic subset that still gives a learner a pass-grade path on the 35-question exam.

Why No Courses

The Extra Bites program needs four longer (30-45 min) Courses because Extra has irreducibly hard topic clusters (resonance and Q, receiver measurement, antenna math, modulation theory) that cannot be honestly taught in a 5-10 minute Bite. General is different:

- Smaller pool (422 questions vs 599 for Extra).
- Less math-heavy. The only deeply math-heavy material on General is the **G5** subelement (reactance, impedance, dB, transformers), which we **reserve** below.
- Most of the pool is rules, operating procedures, propagation, and recognition-level component questions – all of which deliver well as Bite + Dessert content.

So General Bites is **two layers only**: 12 Bites and 12 Desserts. No Courses.

The Two Layers

Layer	Length	Job	Where it lives
General Bite	5-10 min	One compact concept that unlocks several related exam questions. Plain language, one example, exam IDs. May cover one exam group or a natural pair.	Short segment at any monthly club meeting. Handout goes home.
General Dessert	self-study	Rules, privileges, operating procedures, safety – memorization-heavy material that doesn't need live teaching. May cover one exam group or a natural pair.	Distributed alongside Bites and at meetings. Done at home.

Module-to-group ratio: A Bite or Dessert is allowed to own **two related exam groups** when the groups are natural pairs (e.g. G3A propagation basics + G3B HF skip and MUF). This is the same pattern Extra Bites uses for the multi-group Courses (EC-B owns E4B + E4C). It lets a 24-module program (12 Bites + 12 Desserts) cover 30 exam groups.

Pass Math For The General Exam

Quantity	Working assumption
Pool size	422 active questions across 35 exam groups
Exam length	35 questions, one drawn per group
Pass threshold	26 of 35 correct (74%)
Exam Qs on covered material	30 of 35 (the 5 reserved groups still get tested – you just guess)
Accuracy on covered material	about 85%
Guess rate on uncovered material (4-choice)	about 25%
Expected score	30 × 0.85 + 5 × 0.25 ≈ **26.75**

That projects to 27 correct – about one above the 26-pass threshold. The margin is thinner than the Extra Bites design (which targets +1.3 above pass), but General-class learners typically exceed the 85% projection on the easy operating and rules content, so the real-world floor sits closer to 28-30. **If your cohort is shaky and you want more buffer, reserve 6 instead of 5 (drop the projected score to 26.2 with 29 covered).**

Reserved For Separate Self-Study (5 Groups)

These 5 groups are **not covered** by General Bites or General Desserts. They are real topics worth learning – this program leaves them to outside resources because they don't deliver well in 5-10 minute concept segments.

Group	Topic	Qs	Why reserved
G5A	Reactance, inductance, capacitance, impedance	12	Math-heavy; complex-impedance ideas.

G5B	Decibels, current/voltage dividers, electrical power	14	dB and power math is a discipline of its own.
G5C	Resistors, capacitors, inductors in series/parallel; transformers	14	Parallel/series math and transformer ratios.
G7B	Digital circuits; amplifiers; oscillators	11	Abstract digital + analog circuit concepts.
G8C	Digital modes; packet; signal processing terminology	15	Vocabulary cluster; better with a reference.
Total	**Reserved**	**66**	**5 of 35 exam questions**

Alternate scoping: if the cohort is comfortable with dB math (many already-licensed technicians are), swap G5B out and reserve G8A (modulation theory and AM/FM/SSB) or G7C (filters, oscillators, transceiver block) instead. The math-vs-vocabulary balance is a knob the program coordinator turns.

Plan: 12 Bites + 12 Desserts

30 covered groups distributed across 24 modules. Four Bites and two Desserts each own a natural pair of related groups; the rest own one group each.

12 General Bites (one per monthly meeting)

Slot	Title	Groups	Qs
B1	The General-class privileges: what you can do now	G1A	10
B2	Antenna structures, height limits, and FCC notification	G1B	11
B3	Propagation basics: ionosphere, MUF, and HF skip	G3A + G3B	26
B4	Special propagation: scatter, NVIS, polarization	G3C	11
B5	Components as RF parts: passives and ICs	G6A + G6B	23
B6	Power supplies and batteries	G7A	13
B7	The HF transceiver block diagram	G7C	14
B8	Receiver behavior: AGC, IMD, test equipment	G4B	13
B9	Modulation and bandwidth: AM, FM, SSB, sidebands	G8A + G8B	27
B10	Antennas and feedlines: dipoles, verticals, beams	G9A + G9B	23
B11	Vertical antennas and ground systems	G9C	11
B12	Antenna height effects and dipoles	G9D	12
Total	**12 Bites**	**16 groups**	**194**

12 General Desserts (self-study handouts)

ID	Title	Groups	Qs
D1	Safety: RF exposure, AC power, dangerous voltages	G0A + G0B	25
D2	Power limits, prohibited transmissions, identification	G1C	8
D3	VEs, exam administration, and control-operator rules	G1D + G1E	23
D4	Phone operating procedures	G2A	12
D5	Operating courtesies and on-air conduct	G2B	11
D6	CW operating procedures	G2C	11
D7	Volunteer Monitor Program and HF operating	G2D	11
D8	Digital operating: modes, procedures, conventions	G2E	14
D9	Station configuration and operation	G4A	13
D10	RFI to consumer electronics; grounding and bonding	G4C	12
D11	Speech processors, S-meters, sideband near band edges	G4D	11
D12	Mobile and portable HF; alternative energy sources	G4E	11
Total	**12 Desserts**	**14 groups**	**162**

System Totals

Layer	Groups	Pool Qs
Bites	16 (12 modules)	194
Desserts	14 (12 modules)	162
Reserved for separate study	5	66
Pool covered	**30 of 35**	**356 of 422**

Modular Compression

The 12-meeting program length is the **gentlest** setting, not a fixed duration. Every Bite and Dessert is an independent module, so a club or group dials the pace to fit – same materials, same handouts, same recordings, only the schedule changes.

Pace	Bites / mtg	How it runs	Finishes in
Standard	1	One short Bite per monthly meeting. Relaxed; never crowds club business.	12 months
Accelerated Club	2	Two Bites per monthly meeting – only 10-15 min of program time.	6 meetings
Study-Group Sprint	3, weekly	Three Bites per session, run weekly. Still under an hour per session.	4 weeks

The Bites are not mainly a cram tool – their real value is making the next license feel reachable and putting the material in front of a ham early, so a later upgrade is familiar ground instead of first contact. An individual can self-study the same modules at any pace, and VEs or Elmers can hand the whole bundle to an upgrade candidate as a prep packet.

Recommended 12-Meeting Cadence

One Bite per club meeting for 12 months. Desserts go home alongside each Bite, prioritized so rules/safety content comes first (easy points, builds confidence), components and circuits next, antennas at the end.

Meeting	Bite that meeting	Desserts to hand out
1	B1 General privileges	D1 safety; D2 power limits
2	B2 Antenna structures	D3 VEs and control op rules
3	B3 Propagation basics + HF skip	D4 phone procedures
4	B4 Scatter, NVIS, polarization	D5 operating courtesies; D6 CW
5	B5 Components as RF parts	D7 Volunteer Monitor / HF ops
6	B6 Power supplies and batteries	D8 digital operating
7	B7 HF transceiver block diagram	D9 station configuration
8	B8 Receiver behavior and test equipment	D10 RFI and grounding
9	B9 Modulation and bandwidth	D11 speech processors, S-meters
10	B10 Antennas and feedlines	D12 mobile and alternative energy
11	B11 Vertical antennas and ground systems	(catch-up; review weak Desserts)
12	B12 Antenna height effects and dipoles	(final review; mock exam together)

How A Study-Group Sprint Works

A study group is **not a class** – it is three or four people working through the material together, with no teacher. A weekly Sprint session is short and runs itself: play the three Bite recordings for the week (about 15-30 minutes – the recordings do the teaching), then work that week's Dessert sheets together – compare practice-question answers, talk through whatever was confusing, and quiz each other. The whole session stays comfortably under an hour.

Two things keep the format working: **no one has to teach** – the recordings handle that – and **everyone does their own practice-question drill between sessions**, because the group amplifies self-study, it does not replace it.

What The System Will Not Do

- It will not lecture every question in the pool.
- It will not teach the 5 reserved groups (G5A, G5B, G5C, G7B, G8C). Learners who want to master those should use the ARRL General Class License Manual, HamStudy.org explanations, or a dedicated electronics primer.
- It will not replace practice questions. Concept exposure plus repeated practice-question drill is the working combination.

Success Target

Move a Technician from roughly 18 of 35 on a cold General-class practice exam to roughly 28-30 of 35 on a focused exam, where most missed questions are concentrated in the 5 reserved groups. That gives a defensible pass-grade path at whatever pace the club or study group runs.